Compact Modeling Solution of Layout Dependent Effect for FinFET Technology

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Abstract

We successfully developed and verified a complete compact model solution for layout dependent effect (LDE) of FinFET technology. LDE has significant impact on the device performances mainly due to the application of stressors and aggressive device scaling. With LDE, performance degradation may be up to 10% or more. In this work, compact model solution for Length of Oxidation (LOD), Well Proximity Effect (WPE), Neighboring Diffusion Effect (NDE), Metal Boundary Effect (MBE), and Gate Line End Effect (GLE) were delivered. This solution was implemented successfully in BSIM-CMG for efficient circuit simulation.

I. Introduction

In recent industry technology trend, planar CMOS technologies, no matter for Poly SiON or high-K metal gate (HKMG), can be extended to 28nm technology. To continuously follow Morre's law, since 16nm technology and beyond, FinFET technology is becoming the major technology with the aggressive device scaling and excellent performance boost for both logic and mixed-signal applications.. FiNFET devices have the superior immunity of short channel effect due to the effective gate control on the channel. Because this is a 3-dimentional device, total width of a FiNFET device is the sum of fin height and fin thickness, which is more compact than planar CMOS. Fig. 1 shows the schematic of 14nm FinFET devices. Table 1 lists the key features of 14nm FinFET technology. Raised source and drain are used. With the complex schemes, a model for accurate device-level and circuit-level simulations should reflect the LDE behaviors thoroughly. LDE was widely discussed for both planar and FinFET devices due to process complexity and significant impact on device performances in the past years [1]-[3]. Currently, public domain SPICE model like BSIM-CMG provided the robust I-V and C-V model equations for efficient simulation but did not provide the complete LDE solution for FinFET technology. For the first time, a complete compact model solution for LDE was developed and delivered. For FinFET technology, Gate spacing effect (GSE), LOD, WPE, MBE, GLE, and NDE potentially have the significant impact on chip performances due to different layout styles. GSE is due to different gate spacing which results in strain difference in the channel. Because gate pitch selection is very limited in the topology layout rule (TLR), we can set up the flags in the model. There is no need to set up the scalable model equations for GSE. Considering the compatibility and efficient implementation into BSIM-CMG, we set up the proprietary equations based upon BSIM-CMG threshold voltage equations and mobility equations [4]. These LDE model equations were treated as macro model and set up in the model card. Related instance parameters were defined for every LDE and these parameters will be extracted from the GDSII by Layout-Versus-Schematic (LVS) for model simulation.

II. LDE and Compact Model Solution for FinFET Devices

A. Metal Boundary Effect (MBE)

Replacement metal gate (RMG) is widely used in gate-first and gate-last high-K metal gate devices since 28nm CMOS technology. In FinFET technology, RMG was also continuously applied. Dual work function materials to tune the threshold voltage of nMOSFET and pMOSFET result in the proximity effect at the N/P metal gate boundary. This effect is caused by the Al diffusion and residue [3]. IThe impact caused by MBE is on threshold voltage and current of a FiNFET device. The device layout and instance parameters are shown in Fig. 2(a). There are two instance parameters for MBE: MBEYA and MBEYB. The two instance parameters defined the distances from the metal gate boundary to fin edges. Fig. 2(b) shows model equations for MBE to improve the accuracy of threshold voltage, mobility, Vtsat and Eta0. The improved electrical items with MBE modeling are Idsat, Idlin, Vtsat and Vtlin. Fig. 2(c) shows part of the model simulation trends of MBE. Table 2 lists the model parameters of MBE.

B. Gate Line End Effect (GLE)

In FinFET technology, ploy slot cut was used to define a FiNFET device area. Poly gate and spacer were formed first and poly slot cut was used to cut the poly line and define the device area. Then poly gate will be removed and RMG will replace poly gate. If the poly slot cut is very close to the fin edge, quality of RGM filling near the cutting edge will be potentially impacted, which will cause the change of electrical characteristics, like threshold voltage change. GLE layout cell and instance parameters are shown in Fig. 3(a). GLEYA and GLEYB are the two instance parameters of GLE and are defined the spacing from fin edge to the poly slot cut. Fig. 3(b) shows the model equations for GLE. Both MBE and GLE have the similar model equations. The major differences are the naming of these new model parameters. Fig. 3(c) shows the simulation trends of GLE. Table 3 lists the model parameters of GLE.

C. Length of Oxidation (LOD)

For planar devices, LOD effect is mainly caused by Shallow Trench Isolation (STI). In BSIM4, this effect is well modeled and verified by Si. For FinFET device, it is with raised source/drain areas. Usually, SiP and SiGe are used for the raised source and drain areas of nMOSFET and pMOSFET respectively [5]. The stress in the channels of nMOSFET and pMOSFET are mainly from SiP and SiGe [5]. Planar LOD model equations were not suitable and we proposed the model equations for FinFET devices. Device layout and instance parameters of LOD are shown in Fig. 4(a). SA and SB are the instance parameters and are defined as spacing between gate edge and fin edge. Fig. 4(b) shows the model equations of LOD. For both planar devices and FinFET devices, some LOD behaviors are not monotonic; but close to quadratic. The equations inside the dot-line rectangle are the correction items to fit to the quadratic behaviors of LOD. Fig. 4(c) shows the model simulation behaviors of LOD. Table 4 lists the model parameters of LOD.

D. Neighboring Diffusion Effect (NDE)

Aggressive device scaling made the diffusion distance much closer in TLR. Neighboring diffusion area will have the stress on the channel. This phenomenon became significant and needed to model since 40nm technology. The layout and instance parameters are shown in Fig. 5(a). Instance parameters of NDE are NDEXA1, NDEXA2, NDEYA1 and NDEYA2. Instance parameter NDEXA1 is the spacing between two diffusion areas in the X-direction. NEXXA2 is the width of the neighboring diffusion in the

X-direction. NDEYA1 and NDEYA2 are the diffusing spacing of two diffusion areas and width of neighboring diffusion area. NDE model equations in Y-direction are similar to X-direction equations shown in Fig. 5(b). Fig. 5(c) shows the model simulation of NDE trend. Table 5 lists the model parameters of NDE.

E. Well Proximity Effect (WPE)

WPE model was well setup and verified by Si in BSIM4. The mechanism and device behaviors of WPE in FinFET and planar devices are similar. We can directly refer to BSIM4 WPE model. The device layout and instance parameter are shown in Fig. 6(a). When instance parameter SC is small in Fig. 6(b), the combined effect will be WPE and MBE. It is necessary to use another set of WPE-cells to extract the pure WPE. Fig. 6(c) shows the pure WPE simulation results.

III. Verification

To verify the accuracy of the LDE model and check the impact of LDE on circuit level, we used three types of ring oscillators, made of INVERTOR, NAND, and NOR gates to evaluate the performances of tegular-Vt (RVT) devices, low-Vt (LVT) devices and super low Vt (sLVT) devices. Parasitic R/C in the ring oscillators were extracted from GDSII by LPE deck. Table 6 shows the LDE impact on a ring oscillator. Different threshold voltage devices were checked. With LDE, 10% impact on speed was observed.

VI. Conclusion

A complete set of FinFET LDE modeling solution was provided. These LDE model equations were successfully verified and implemented and the instance parameters were set up in the verified LVS deck. RO simulation was demonstrated. This solution is used for 14nm FinFET model delivery already and can be applied further for 10nm FinFET technology.

Acknowledgment

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References

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[3] S. Sato, et al., Symp. VLSI Tech. Dig., pp. 116 – 117, 2013.
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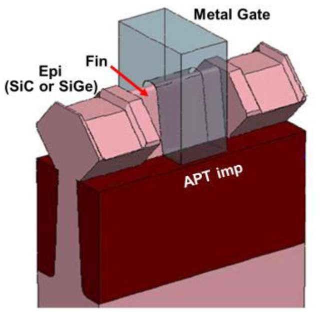
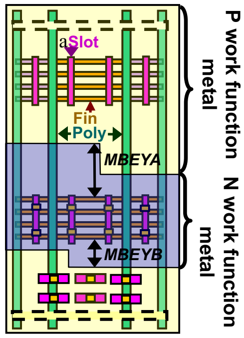


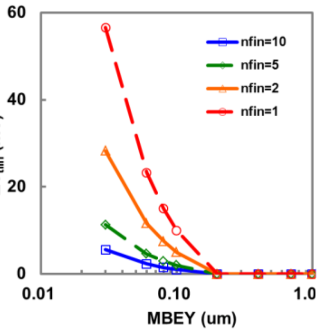
Fig. 1 Schematic of 14nm FinFET structure.

Table 1. Key modules of 14FF+

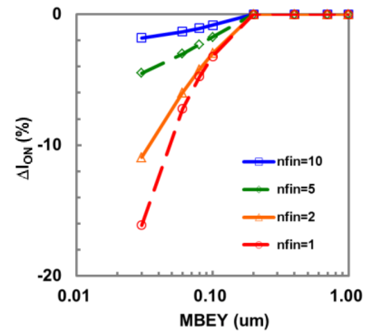
Key Features	**14FF+**
Isolation	**Fin with single STI**
Gate Dielectric	**HFO-based HK**
Gate	**Dual MG, HK Last**
Spacer	**Lower k Spacer**
Stress Engineering	**eSiP(nMOS) / eSiGe(pMOS)**
Salicide	**TiSi**
Litho	**Immersion**
BEOL Metal	**Co liner/Cu**
BEOL Dielectric	**AlN/NDC/ULK**



2(a)



2(c)



2(d)

$$DVTSHIFT = DVTSHIFT_{original} + \frac{KMDVTSHIFT}{KMstress\_DVTSHIFT}\cdot(Inv\_\mathrm{mbeya} + Inv\_\mathrm{mbeyb} - Inv\_\mathrm{mbeya}_{ref} - Inv\_\mathrm{mbeyb}_{ref})*\left(\frac{1}{NFIN}\right)^{\mathrm{KMNFINFACTOR}}$$

$$ETA0 = ETA0_{original} + \frac{KMETA\_TOTAL}{KMstress\_DVTSHIFT^{LODKMETA0}}\cdot(Inv\_\mathrm{mbeya} + Inv\_\mathrm{mbeyb} - Inv\_\mathrm{mbeya}_{ref} - Inv\_\mathrm{mbeyb}_{ref})*\left(\frac{1}{NFIN}\right)^{\mathrm{KMNFINFACTOR}}$$

$$KMETA\_TOTAL = KMETA0 + \frac{LKMETA0}{(L_{drawn}+XL)} + \frac{NKMETA0}{NFIN} + \frac{PKMETA0}{(L_{drawn}+XL)\cdot NFIN}$$

where $Inv\_\mathrm{mbeya}_{ref} = \frac{1}{MBEYAREF}$, $Inv\_\mathrm{mbeyb}_{ref} = \frac{1}{MBEYBREF}$, $Inv\_\mathrm{mbeya} = \frac{1}{MBEYA1}$, $Inv\_\mathrm{mbeyb} = \frac{1}{MBEYB1}$,

$$KMstress\_DVTSHIFT = 1 + \frac{LKMDVTSHIFT}{(L_{drawn}+XL)^{LLODKMVTH}} + \frac{NKMDVTSHIFT}{NFIN^{NLODKMVTH}} + \frac{PKMDVTSHIFT}{(L_{drawn}+XL)^{LLODKMVTH}\cdot NFIN^{NLODKMVTH}}$$

2(b)

$$\rho_{\mu eff}(MBEYA, MBEYB) = \frac{KMU0}{KMstress\_\mu 0}\cdot(Inv\_mbeya + Inv\_mbeyb)$$

where $KMstress\_\mu 0 = \left(1 + \frac{LKMU0}{(L_{drawn}+XL)^{LLODKMU0}} + \frac{NKMU0}{NFIN^{NLODKMU0}} + \frac{PKMU0}{(L_{drawn}+XL)^{LLODKMU0}\cdot NFIN^{NLODKMU0}}\right)\times\left(1 + TKMU0\cdot\left(\frac{Temperature}{TNOM} - 1\right)\right)$

so that $\mu_{eff} = \mu_{eff0}\cdot\left(1 + \rho_{\mu eff}(MBEYA, MBEYB)*\left(\frac{1}{NFIN}\right)^{\mathrm{KMNFINFACTOR}}\right)\cdot\left(1 + \rho_{\mu eff}(MBEYAREF, MBEYBREF)*\left(\frac{1}{NFIN}\right)^{\mathrm{KMNFINFACTOR}}\right)^{-1}$

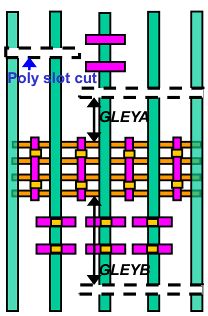
$$\upsilon_{sat} = \upsilon_{sat0}\left(1 + KMVSAT\_TOTAL\cdot\rho_{\mu eff}(MBEYA, MBEYB)\cdot\left(\frac{1}{NFIN}\right)^{\mathrm{KMNFINFACTOR}}\right)\cdot\left(1 + KMVSAT\_TOTAL\cdot\rho_{\mu eff}(MBEYAREF, MBEYBREF)\cdot\left(\frac{1}{NFIN}\right)^{\mathrm{KMNFINFACTOR}}\right)^{-1}$$

$\mu_{eff0}, \upsilon_{sat0}$ are low field mobility and saturation velocity at $MBEYAREF, MBEYBREF$.

where $KMVSAT\_TOTAL = KMVSAT + \frac{LKMVSAT}{(L_{drawn}+XL)} + \frac{NKMVSAT}{NFIN} + \frac{PKMVSAT}{(L_{drawn}+XL)\cdot NFIN}$

2(b)

Fig. 2(a) Schematic illustration of device layout and instance parameters of MBE, **(b)** threshold and mobility related model equations for describing MBE, **(c)** and **(d)** MBE test on nMOSFET with the channel length of 0.02um.



3(a)

$$DVTSHIFT = DVTSHIFT_{original} + \frac{KGDVTSHIFT}{KGstress\_DVTSHIFT}\cdot(Inv\_\mathrm{gleya} + Inv\_\mathrm{gleyb} - Inv\_\mathrm{gleya}_{ref} - Inv\_\mathrm{gleyb}_{ref})*\left(\frac{1}{NFIN}\right)^{\mathrm{KGNFINFACTOR}}$$

$$ETA0 = ETA0_{original} + \frac{KGETA\_TOTAL}{KGstress\_DVTSHIFT^{LODKGETA0}}\cdot(Inv\_\mathrm{gleya} + Inv\_\mathrm{gleyb} - Inv\_\mathrm{gleya}_{ref} - Inv\_\mathrm{gleyb}_{ref})*\left(\frac{1}{NFIN}\right)^{\mathrm{KGNFINFACTOR}}$$

$$KGETA\_TOTAL = KGETA0 + \frac{LKGETA0}{(L_{drawn}+XL)} + \frac{NKGETA0}{NFIN} + \frac{PKGETA0}{(L_{drawn}+XL)\cdot NFIN},$$

where $Inv\_gleya_{ref} = \frac{1}{GLEYAREF}$, $Inv\_gleyb_{ref} = \frac{1}{GLEYBREF}$, $Inv\_gleya = \frac{1}{GLEYA1}$, $Inv\_gleyb = \frac{1}{GLEYB1}$,

$$KGstress\_DVTSHIFT = 1 + \frac{LKGDVTSHIFT}{(L_{drawn}+XL)^{LLODKGVTH}} + \frac{NKGDVTSHIFT}{NFIN^{NLODKGVTH}} + \frac{PKGDVTSHIFT}{(L_{drawn}+XL)^{LLODKGVTH}\cdot NFIN^{NLODKGVTH}}$$

3(b)

$$\rho_{\mu eff}(GLEYA, GLEYB) = \frac{KGU0}{KGstress\_\mu 0}\cdot(Inv\_gleya + Inv\_gleyb)$$

where $KGstress\_\mu 0 = \left(1 + \frac{LKGU0}{(L_{drawn}+XL)^{LLODKGU0}} + \frac{NKGU0}{NFIN^{NLODKGU0}} + \frac{PKGU0}{(L_{drawn}+XL)^{LLODKGU0}\cdot NFIN^{NLODKGU0}}\right)\cdot\left(1 + TKGU0\cdot\left(\frac{Temperature}{TNOM} - 1\right)\right)$

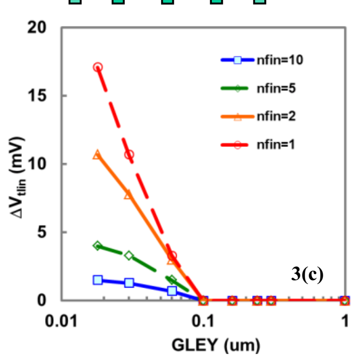
so that $\mu_{eff} = \mu_{eff0}\cdot\left(1 + \rho_{\mu eff}(GLEYA, GLEYB)*\left(\frac{1}{NFIN}\right)^{\mathrm{KGNFINFACTOR}}\right)\cdot\left(1 + \rho_{\mu eff}(GLEYAREF, GLEYBREF)*\left(\frac{1}{NFIN}\right)^{\mathrm{KGNFINFACTOR}}\right)^{-1}$,

$$\upsilon_{sat} = \upsilon_{sat0}\cdot\left(1 + KGVSAT\_TOTAL\cdot\rho_{\mu eff}(GLEYA, GLEYB)*\left(\frac{1}{NFIN}\right)^{\mathrm{KGNFINFACTOR}}\right)\cdot\left(1 + KGVSAT\_TOTAL\cdot\rho_{\mu eff}(GLEYAREF, GLEYBREF)*\left(\frac{1}{NFIN}\right)^{\mathrm{KGNFINFACTOR}}\right)^{-1}$$

$\mu_{eff0}, \upsilon_{sat0}$ are low field mobility and saturation velocity at $GLEYAREF, GLEYBREF$.

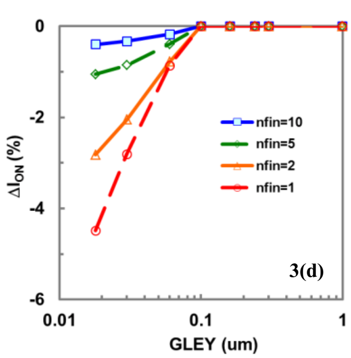
where $KGVSAT\_TOTAL = KGVSAT + \frac{LKGVSAT}{(L_{drawn}+XL)} + \frac{NKGVSAT}{NFIN} + \frac{PKGVSAT}{(L_{drawn}+XL)\cdot NFIN}$

3(b)

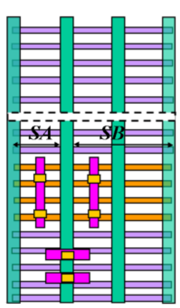


3(c)

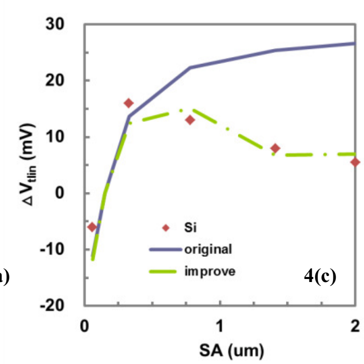
Fig. 3(a) Schematic illustration of device layout and instance parameters of GLE, **(b)** threshold and mobility related model equations for describing GLE, **(c)** and **(d)** GLE test on nMOSFET with the channel length of 0.02um..



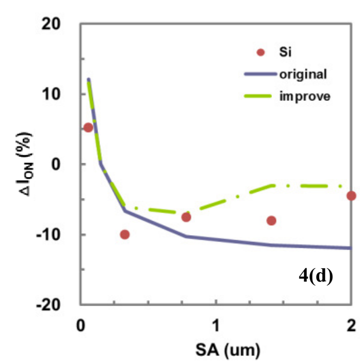
3(d)



4(a)



4(c)



4(d)

$$DVTSHIFT = DVTSHIFT_{original} + \frac{KVTH0}{Kstress\_vth0}\cdot(Inv\_sa + Inv\_sb - Inv\_sa_{ref} - Inv\_sb_{ref} + STIVTH0\cdot(Inv\_sa^2 + Inv\_sb^2 - Inv\_sa_{ref}{}^2 - Inv\_sb_{ref}{}^2)) + \frac{KVTH1}{Kstress\_vth1}\cdot(isa1_{eff} + isb1_{eff} - isa1_{ref} - isb1_{ref})$$

$$K2 = K2_{original} + \frac{STK2}{Kstress\_vth0^{LODK2}}\cdot(Inv\_sa + Inv\_sb - Inv\_sa_{ref} - Inv\_sb_{ref})$$

$$ETA0 = ETA0_{original} + \frac{STETA0}{Kstress\_vth0^{LODKETA0}}\cdot(Inv\_sa + Inv\_sb - Inv\_sa_{ref} - Inv\_sb_{ref} + STIETA0\cdot(Inv\_sa^2 + Inv\_sb^2 - Inv\_sa_{ref}{}^2 - Inv\_sb_{ref}{}^2))$$

where $Inv\_sa_{ref} = (SAREF + XL + 0.5\cdot L_{drawn})^{-1}$, $Inv\_sgb_{ref} = (SBREF + XL + 0.5\cdot L_{drawn})^{-1}$, $Inv\_sga = (SA + XL + 0.5\cdot L_{drawn})^{-1}$, $Inv\_sgb = (SB + XL + 0.5\cdot L_{drawn})^{-1}$

$$Kstress\_vth0 = 1 + \frac{LKVTH0}{(L+XL)^{LLODKVTH}} + \frac{WKVTH0}{(NFIN)^{WLODKVTH}} + \frac{PKVTH0}{(L+XL)^{LLODKVTH}\cdot(NFIN)^{WLODKVTH}},$$

$$Kstress\_vth1 = 1 + \frac{LKVTH1}{(L+XL)^{LLODKVTH1}} + \frac{WKVTH1}{(NFIN)^{WLODKVTH1}} + \frac{PKVTH1}{(L+XL)^{LLODKVTH1}\cdot(NFIN)^{WLODKVTH1}},$$

$$isa1_{eff} = \left(1 + \left(\frac{LOD1C}{SA + 0.5\cdot L_{drawn}}\right)^{LOD1P}\right)^{-1},\ isb1_{eff} = \left(1 + \left(\frac{LOD1C}{SB + 0.5\cdot L_{drawn}}\right)^{LOD1P}\right)^{-1},\ isa1_{ref} = \left(1 + \left(\frac{LOD1C}{SAREF + 0.5\cdot L_{drawn}}\right)^{LOD1P}\right)^{-1},\ isb1_{ref} = \left(1 + \left(\frac{LOD1C}{SBREF + 0.5\cdot L_{drawn}}\right)^{LOD1P}\right)^{-1}$$

4(b)

$$\rho_{\mu eff}(SA,SB) = \frac{KU0}{Kstress\_\mu 0}\cdot\left(Inv\_sa + Inv\_sb + STIU0\cdot\left(Inv\_sa^2 + Inv\_sb^2\right)\right),\ \boxed{\rho_{\mu eff1}(SA,SB) = \frac{KU1}{Kstress\_\mu 1}\cdot\left(isa_{eff} + isb_{eff}\right)}$$

$$\rho_{vsat}(SA,SB) = \frac{KU0}{Kstress\_\mu 0}\cdot\left(Inv\_sa + Inv\_sb + STIVSAT\cdot\left(Inv\_sa^2 + Inv\_sb^2\right)\right)$$

where $Kstress\_\mu 0 = \left(1 + \frac{LKU0}{(L_{drawn}+XL)^{LLODKU0}} + \frac{WKU0}{(NFIN)^{WLODKU0}} + \frac{PKU0}{(L_{drawn}+XL)^{LLODKU0}\cdot(NFIN)^{WLODKU0}}\right)\times\left(1 + TKU0\cdot\left(\frac{Temperature}{TNOM} - 1\right)\right),$

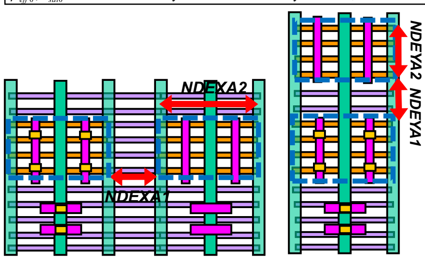
$$\boxed{Kstress\_\mu 1 = \left(1 + \frac{LKU1}{(L_{drawn}+XL)^{LLODKU1}} + \frac{WKU1}{(NFIN)^{WLODKU1}} + \frac{PKU1}{(L_{drawn}+XL)^{LLODKU1}\cdot(NFIN)^{WLODKU1}}\right)\times\left(1 + TKU1\cdot\left(\frac{Temperature}{TNOM} - 1\right)\right),}$$

so that $\mu_{eff} = \mu_{eff0}\cdot\frac{1+\rho_{\mu eff}(SA,SB)}{1+\rho_{\mu eff}(SAREF,SBREF)}\frac{1+\rho_{\mu eff1}(SA,SB)}{1+\rho_{\mu eff1}(SAREF,SBREF)},\ \upsilon_{sat} = \upsilon_{sat0}\cdot\frac{1+KVSAT\cdot\rho_{vsat}(SA,SB)}{1+KVSAT\cdot\rho_{vsat}(SAREF,SBREF)}$

$\mu_{eff0}, \upsilon_{sat0}$ are low field mobility and saturation velocity at $SAREF$, $SBREF$.

4(b)

Fig. 4(a) Schematic illustration of device layout and instance parameters of LOD effect, **(b)** threshold and mobility related model equations for describing LOD effect, **(c)** and **(d)** LOD effect on nMOSFET considering its correction terms on nMOSFET with channel length of 0.02um, nfin=4.



5(a)

$$DVTSHIFT = DVTSHIFT_{original} + \frac{NDEXAVTH0}{NDEXAstress\_vth0}\cdot\left(Inv\_ndexa_1 + Inv\_ndexa_2\right) - \frac{NDEXAVTH0_{NNN/PPP}}{NDEXAstress\_vth0}\cdot\left(Inv\_ndexa_{1ref} + Inv\_ndexa_{2ref}\right)$$

$$ETA0 = ETA0_{original} + \frac{NDEXAETA}{NDEXAstress\_vth0^{LODNDEXAETA0}}\cdot\left(Inv\_ndexa_1 + Inv\_ndexa_2\right) - \frac{NDEXAETA_{NNN/PPP}}{NDEXAstress\_vth0^{LODNDEXAETA0}}\cdot\left(Inv\_ndexa_{1ref} + Inv\_ndexa_{2ref}\right)$$

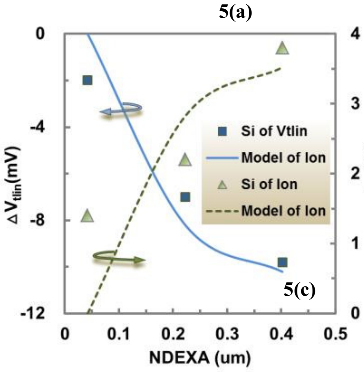
$$NDEXAETA = NDEXAETA0 + \frac{LNDEXAETA0}{(L_{drawn}+XL)} + \frac{WNDEXAETA0}{(NFIN)} + \frac{PNDEXAETA0}{(L_{drawn}+XL)\cdot(NFIN)}$$

where $Inv\_ndexa_{1ref} = \left(NDEXAREF + SAREF + XL + 0.5\cdot L_{drawn}\right)^{-1}$, $Inv\_ndexa_{2ref} = \left(NDEXAREF + SBREF + XL + 0.5\cdot L_{drawn}\right)^{-1}$

$Inv\_ndexa_1 = \left(NDEXA1 + SA + 0.5\cdot L\right)^{-1}$, $Inv\_ndexa_2 = \left(NDEXA2 + SB + 0.5\cdot L\right)^{-1}$

$$\mathrm{NDEXA}stress\_vth0 = 1 + \frac{LNDEXATH0}{(L_{drawn}+XL)^{LLODNDEXAVTH}} + \frac{WNDEXAVTH0}{(NFIN)^{WLODNDEXAVTH}} + \frac{PNDEXAVTH0}{(L_{drawn}+XL)^{LLODNDEXAVTH}\cdot(NFIN)^{WLODNDEXAVTH}}$$

5(b)



$$\rho_{\mu eff}(NDEXA1, NDEXA2) = \frac{NDEXAU0}{NDEXAstress\_\mu 0}\cdot(Inv\_ndexa_1 + Inv\_ndexa_2)$$

$$NDEXAVSATotal = \left(NDEXAVSAT + \frac{LNDEXAVSAT}{(L_{drawn}+XL)} + \frac{WNDEXAVSAT}{(NFIN)} + \frac{PNDEXAVSAT}{(L_{drawn}+XL)\cdot(NFIN)}\right)$$

where

$$\mathrm{NDEXA}stress\_\mu 0 = \left(1 + \frac{LNDEXAU0}{(L_{drawn}+XL)^{LLODNDEXAU0}} + \frac{WNDEXAU0}{(NFIN)^{WLODNDEXAU0}} + \frac{PNDEXAU0}{(L_{drawn}+XL)^{LLODNDEXAU0}\cdot(NFIN)^{WLODNDEXAU0}}\right)\times\left(1 + TNDEXAU0\cdot\left(\frac{Temperature}{TNOM} - 1\right)\right)$$

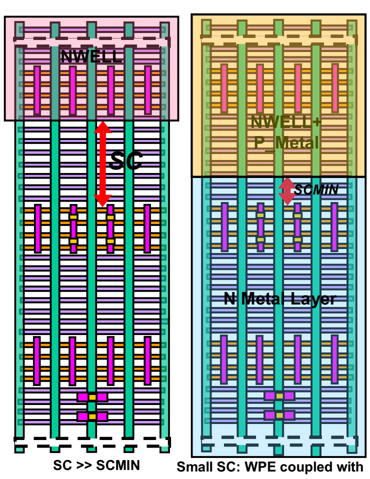
so that $\mu_{eff} = \mu_{eff0}\cdot\left(1 + \rho_{\mu eff}(NDEXA1, NDEXA2)\right)\cdot\left(1 + \rho_{\mu eff}(NDEXAREF, NDEXAREF)_{NNN/PPP}\right)^{-1}$

$\upsilon_{sat} = \upsilon_{sat0}\cdot\left(1 + NDEXAVSATotal\cdot\rho_{\mu eff}(NDEXA1, NDEXA2)\right)\cdot\left(1 + NDEXAVSATotal\cdot\rho_{\mu eff}(NDEXAREF, NDEXAREF)_{NNN/PPP}\right)^{-1}$

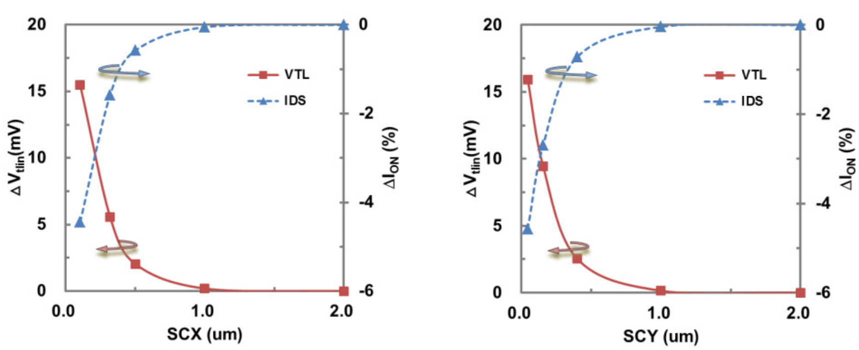
$\mu_{eff0}, \upsilon_{sat0}$ are low field mobility and saturation velocity at $NDEXAREF$.

5(b)

Fig. 5(a) Schematic illustration of device layout and instance parameters of NDE, **(b)** threshold and mobility related model equations for describing NDE, **(c)** modeling result of NDE on nMOSFET with the channel length of 0.02um, nfin=4.



6(a) 6(b)



6(c) 6(d)

Fig. 6(a) Schematic illustration of device layout and instance parameters of WPE, **(b)** verification layout for WPE, **(c)** and **(d)** characteristics of threshold voltage shift vs. SCX and SCY.

Table 2. Parameters for modeling MBE.

Model Parameter

Parameter	Description	Default	Limition	Unit	Binnable
MBEYAREF	Reference deistance for MBEYA	1.00E-06	>0	m	No
MBEYBREF	Reference deistance for MBEYB	1.00E-06	>0	m	No
KMDVTSHIFT	Threshold shift parameter for Metal Boundary Effect	0	No	V*m	No
LKMDVTSHIFT	Length dependence of KMDVTSHIFT	0	No	none	No
NKMDVTSHIFT	NFIN dependence of KMDVTSHIFT	0	No	none	No
PKMDVTSHIFT	Cross-term dependence of KMDVTSHIFT	0	No	none	No
LLODKMVTH	Power-order of length dependence of KMDVTSHIFT	0	>0	none	No
NLODKMVTH	Power-order of NFIN dependence of KMDVTSHIFT	0	>0	none	No
KMETA0	ETA0 shift parameter for Metal Boundary Effect	0	No	m	Yes
LKMETA0	Length binning term of KMETA0	0	No	none	No
NKMETA0	NFIN binning term of KMETA0	0	No	none	No
PKMETA0	Product binning term of KMETA0	0	No	none	No
LODKMETA0	ETA0 shift modification factor for power-order parameter of kmdvtshift_stress	0	>0	none	No
KMU0	Mobility degradation/enhancement coefficient for Metal Boundary Effect	0	No	m	No
LKMU0	Length dependence of KMU0	0	No	none	No
NKMU0	NFIN dependence of KMU0	0	No	none	No
PKMU0	Cross-term dependence of KMU0	0	No	none	No
LLODKMU0	Power-order of length dependence of KMU0	0	>0	none	No
NLODKMU0	Power-order of NFIN dependence of KMU0	0	>0	none	No
TKMU0	Temperature coefficient of KMU0	0	No	none	No
KMVSAT	Saturation velocity degradation/ enhancement parameter for Metal Boundary Effect	0	No	m	Yes
LKMVSAT	Length binning term of KMVSAT	0	No	none	No
NKMVSAT	NFIN binning term of KMVSAT	0	No	none	No
PKMVSAT	Product binning term of KMVSAT	0	No	none	No
KMNFINFACTOR	Power-order parameter of NFIN	0	>=0	none	No

Table 3. Parameters for modeling GLE

Model Parameter

Parameter	Description	Default	Limition	Unit	Binnable
GLEYAREF	Reference deistance for GLEYA	1.00E-06	>0	m	No
GLEYBREF	Reference deistance for GLEYB	1.00E-06	>0	m	No
KGDVTSHIFT	Threshold shift parameter for Gate-Line End Effect	0	No	V*m	No
LKGDVTSHIFT	Length dependence of KGDVTSHIFT	0	No	none	No
NKGDVTSHIFT	NFIN dependence of KGDVTSHIFT	0	No	none	No
PKGDVTSHIFT	Cross-term dependence of KGDVTSHIFT	0	No	none	No
LLODKGVTH	Power-order of length dependence of KGDVTSHIFT	0	>0	none	No
NLODKGVTH	Power-order of NFIN dependence of KGDVTSHIFT	0	>0	none	No
KGETA0	ETA0 shift parameter for Gate-Line End Effect	0	No	m	Yes
LKGETA0	Length binning term of KGETA0	0	No	none	No
NKGETA0	NFIN binning term of KGETA0	0	No	none	No
PKGETA0	Product binning term of KGETA0	0	No	none	No
LODKGETA0	ETA0 shift modification factor for power-order parameter of kgdvtshift_stress	0	>0	none	No
KGU0	Mobility degradation/enhancement coefficient for Gate-Line End Effect	0	No	m	No
LKGU0	Length dependence of KGU0	0	No	none	No
NKGU0	NFIN dependence of KGU0	0	No	none	No
PKGU0	Cross-term dependence of KGU0	0	No	none	No
LLODKGU0	Power-order of length dependence of KGU0	0	>0	none	No
NLODKGU0	Power-order of NFIN dependence of KGU0	0	>0	none	No
TKGU0	Temperature coefficient of KGU0	0	No	none	No
KGVSAT	Saturation velocity degradation/ enhancement parameter for Gate-Line End Effect	0	No	m	Yes
LKGVSAT	Length binning term of KGVSAT	0	No	none	No
NKGVSAT	NFIN binning term of KGVSAT	0	No	none	No
PKGVSAT	Product binning term of KGVSAT	0	No	none	No
KGNFINFACTOR	Power-order parameter of NFIN	0	>=0	none	No

Table 4. Parameters for modeling LOD

Model Parameter

Parameter	Description	Default	Limitation	Unit	Binnable
SAREF	Reference distance for SA, >0.0	by technology	>0	m	No
SBREF	Reference distance for SB, >0.0	by technology	>0	m	No
WLOD	Width parameter for stress effect	0	none	m	No
KVTH0	Threshold shift parameter for stress effect	0	none	V*m	No
LKVTH0	Length dependence of KVTH0	0	none	none	No
WKVTH0	Width dependence of KVTH0	0	none	none	No
PKVTH0	Product dependence of KVTH0	0	none	none	No
LLODKVTH	Power-order of length dependence of KVTH0	0	none	none	No
WLODKVTH	Power-order of width dependence of KVTH0	0	none	none	No
STETA0	ETA0 shift factor for LOD	0	none	none	No
LODKETA0	Power-order parameter of ETA0 factor modification	1	none	none	No
STK2	K2 shift factor for LOD	0	none	none	No
LODK2	Power-order parameter of K2 factor modification	1	none	none	No

Table 4. Parameters for modeling LOD (continue)

KU0	Mobility degradation/enhancement coefficient for poly pitch stress effect	0	none	m	No
LKU0	Length binning term of KU0	0	none	none	No
WKU0	Width binning term of KU0	0	none	none	No
PKU0	Product binning term of KU0	0	none	none	No
LLODKU0	Power order of length dependence of KU0	0	none	none	No
WLODKU0	Power order of width dependence of KU0	0	none	none	No
TKU0	Temperature coefficient of KU0	0	none	none	No
KVSAT	Saturation velocity degradation/enhancement for poly picth stress effect	0	none	m	No
STIVTH0	Secondary VTH0 parameter	0	none	none	No
STIETA0	Secondary ETA0 parameter	0	none	none	No
STIU0	Secondary U0 parameter	0	none	none	No
STIVSAT	Secondary VSAT parameter	0	none	none	No
KVTH1	Threshold shift parameter for stress effect	0	none	V*m	No
LKVTH1	Length dependence of KVTH1	0	none	none	No
WKVTH1	Width dependence of KVTH1	0	none	none	No
PKVTH1	Product dependence of KVTH1	0	none	none	No
LOD1C	Length parameter for stress effect	0	none	m	No
LOD1P	Power-order of length dependence of LOD1C	0	none	none	No
KU1	Mobility degradation/enhancement coefficient for poly pitch stress effect	0	none	m	No
LKU1	Length binning term of KU1	0	none	none	No
WKU1	Width binning term of KU1	0	none	none	No
PKU1	Product binning term of KU1	0	none	none	No
LLODKU1	Power order of length dependence of KU1	0	none	none	No
WLODKU1	Power order of width dependence of KU1	0	none	none	No
TKU1	Temperature coefficient of KU1	0	none	none	No

Table 5. Parameters for modeling NDE.

Model Parameter

Parameter	Description	Default	Limitation	Unit	Binnable
NDEX(/Y)AREF	Reference deistance for NDEX(/Y)A1 and NDEX(/Y)A2	by technology	>0	m	No
NDEX(/Y)AVTH0	Threshold voltage shift parameter for poly pitch stress effect	0	none	V*m	No
LNDEX(/Y)AVTH0	Length binning term of NDEX(/Y)AVTH0	0	none	none	No
WNDEX(/Y)AVTH0	Width binning term of NDEX(/Y)AVTH0	0	none	none	No
PNDEX(/Y)AVTH0	Product binning term of NDEX(/Y)AVTH0	0	none	none	No
LLODNDEX(/Y)AVTH	Power-order of length dependence of NDEX(/Y)AVTH0	0	none	none	No
WLODNDEX(/Y)AVTH	Power-order of width dependence of NDEX(/Y)AVTH0	0	none	none	No
NDEX(/Y)AETA0	ETA0 shift factor for poly pitch stress effect	0	none	none	No
LNDEX(/Y)AETA0	Length binning term of NDEX(/Y)AETA0	0	none	none	No
WNDEX(/Y)AETA0	Width binning term of NDEX(/Y)AETA0	0	none	none	No
PNDEX(/Y)AETA0	Product binning term of NDEX(/Y)AETA0	0	none	none	No
LODNDEX(/Y)AETA0	Power-order parameter of NDEX(/Y)AETA0 factor modification	1	none	none	No
NDEX(/Y)AU0	Mobility degradation/enhancement coefficient for poly pitch stress effect	0	none	m	No
LNDEX(/Y)AU0	Length binning term of NDEX(/Y)AU0	0	none	none	No
WNDEX(/Y)A(/B)U0	Width binning term of NDEX(/Y)AU0	0	none	none	No
PNDEX(/Y)A(/B)U0	Product binning term of NDEX(/Y)AU0	0	none	none	No
LLODNDEX(/Y)A(/B)U0	Power order of length dependence of NDEX(/Y)AU0	0	none	none	No
WLODNDEX(/Y)A(/B)U0	Power order of width dependence of NDEX(/Y)AU0	0	none	none	No
TNDEX(/Y)A(/B)U0	Temperature coefficient of NDEX(/Y)AU0	0	none	none	No
NDEX(/Y)A(/B)VSAT	Saturation velocity degradation/enhancement for poly picth stress effect	0	none	m	No
LNDEX(/Y)A(/B)VSAT	Length binning term of NDEX(/Y)AVSAT	0	none	m	No
WNDEX(/Y)A(/B)VSAT	Width binning term of NDEX(/Y)AVSAT	0	none	m	No
PNDEX(/Y)A(/B)VSAT	Product binning term of NDEX(/Y)AVSAT	0	none	m	No

Table 6. Evaluation of regular, low Vt, and super-low Vt devices with and without LDE.

14FinFET RO w/i wire load	RVT	LVT	sLV
Normalized Freq. w/i LDE	1.00	1.35	1.69
Normalized Freq. w/o LDE	1.09	1.49	1.86
Si	1.01	1.34	1.70